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Fundamentals of Statistical Learning

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Crash Course

1.1 Why Probability?

Observed data are the results of processes whose behavior we may not fully know. To reason about them, we describe the process with a probabilistic model. Probability lets us work forward from a model to possible outcomes; statistical inference uses observed outcomes to learn about the model.

Example 1: A coin toss

Let $\theta = P(\text{heads})$ be the probability that a coin lands heads. If θ is known, we can ask how likely it is to observe three heads in five tosses. If instead we observe three heads in five tosses while θ is unknown, we can use those data to learn about θ .

1.2 Experiments, Outcomes, and Events

Before assigning probabilities, we must specify what the experiment can produce. We start with a finite set of possible results.

Definition 1: Outcomes and events

A random experiment is a process whose result is not known in advance. Each possible result is an *outcome* ω . The set of all possible outcomes is the *sample space* Ω . An *event* is a subset $A \subseteq \Omega$; it occurs when the observed outcome belongs to A .

Example 2: Rolling a die

For one roll of a six-sided die, the sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event “the result is even” is $A = \{2, 4, 6\}$. The whole sample space Ω is the certain event, while $\emptyset$ is the impossible event.

1.3 Operations on Events

Events are sets, so set operations let us describe more complex events. For $A, B \subseteq \Omega$, the union $A \cup B$ occurs when at least one of A and B occurs. The intersection $A \cap B$ occurs when both occur. The complement $A^c = \Omega \setminus A$ occurs when A does not. Two events are *disjoint* if $A \cap B = \emptyset$. If $A \subseteq B$, every occurrence of A also makes B occur.

Example 3: Combining die events

Keep $A = \{2, 4, 6\}$ from Example 2, and let $B = \{4, 5, 6\}$ be the event “the result is at least four”. Then

$$A \cup B = \{2, 4, 5, 6\}, \quad A \cap B = \{4, 6\}, \quad A^c = \{1, 3, 5\}.$$

Proposition 1: De Morgan’s laws

For any events A and B ,

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

In words, *neither A nor B* means that both A and B fail. Likewise, *not both A and B* means that at least one of them fails. The first statement turns a union into an intersection; the second turns an intersection into a union.

Proof. Fix an outcome $\omega \in \Omega$. For the first identity, check when ω belongs to the left-hand side:

$$\begin{aligned} \omega \in (A \cup B)^c &\iff \omega \notin A \cup B \\ &\iff \omega \notin A \text{ and } \omega \notin B \\ &\iff \omega \in A^c \cap B^c. \end{aligned}$$

The same reasoning for the second identity gives

$$\begin{aligned} \omega \in (A \cap B)^c &\iff \omega \notin A \cap B \\ &\iff \omega \notin A \text{ or } \omega \notin B \\ &\iff \omega \in A^c \cup B^c. \end{aligned}$$

Each pair of sets has the same members, so the identities follow. $\square$

For the die events above, $A \cap B = \{4, 6\}$, so $(A \cap B)^c = \{1, 2, 3, 5\}$. We get the same set by taking $A^c \cup B^c = \{1, 3, 5\} \cup \{1, 2, 3\}$.

The same logic applies to any finite collection of events:

$$\left(\bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c, \quad \left(\bigcap_{i=1}^n A_i \right)^c = \bigcup_{i=1}^n A_i^c.$$

For example, an outcome lies outside $\bigcap_i A_i$ exactly when it fails to belong to at least one A_i ; that is precisely when it belongs to $\bigcup_i A_i^c$. This is the version used later in the proof of Bonferroni’s inequality.

1.4 Decomposing Events

A useful way to study an event is to split it into disjoint pieces. For any events A and B , exactly one of B and B^c occurs, so

$$A = A \cap \Omega = A \cap (B \cup B^c) = (A \cap B) \cup (A \cap B^c).$$

The two events on the right are disjoint. For example, a car can fail an emissions test either while it pollutes or while it does not.

More generally, a *partition* $B_1, \dots, B_n$ of Ω consists of pairwise disjoint events whose union is Ω . Every event A can then be written as

$$A = \bigcup_{i=1}^n (A \cap B_i), \quad (A \cap B_i) \cap (A \cap B_j) = \emptyset \quad (i \neq j).$$

This decomposition will let us add the probabilities of different ways in which the same event can occur.

1.5 Assigning Probabilities

There are several ways to interpret a probability. Under a *symmetry* assumption, equally likely outcomes give $P(A) = |A|/|\Omega|$. A *frequency* interpretation concerns the long-run proportion of repetitions in which A occurs. A *subjective* interpretation expresses a degree of belief about A . The axioms below state the mathematical rules that a probability assignment must satisfy, regardless of its interpretation.

Example 4: Two dice

If the 36 ordered results of rolling two dice are equally likely, they form the sample space $\Omega = \{1, \dots, 6\}^2$. The sum is eight for $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$, so the probability of that event is $5/36$. Counting outcomes this way is valid because the ordered pairs are equally likely.

Definition 2: Probability on a finite sample space

A probability measure P assigns a number $P(A)$ to every event $A \subseteq \Omega$ such that:

1. $0 \leq P(A) \leq 1$ for every event A ;
2. $P(\Omega) = 1$;
3. if $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

The last property is called *finite additivity*.

Because A and A^c are disjoint and their union is Ω , these axioms give $P(A^c) = 1 - P(A)$. In particular, $P(\emptyset) = 0$.

They also imply *monotonicity*: if $A \subseteq B$, then $B = A \cup (B \cap A^c)$ is a disjoint union, so $P(B) = P(A) + P(B \cap A^c) \geq P(A)$.

Example 5: A fair die

If the six outcomes of a die roll are equally likely, each has probability $1/6$. The event $A = \{2, 4, 6\}$ therefore has probability $P(A) = 3/6 = 1/2$.

For an infinite sample space, finite additivity is replaced by *countable additivity*: for pairwise disjoint events $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

In that setting, a probability measure is defined on a suitable family of events rather than necessarily on every subset of Ω . On a finite sample space, the finite formulation above is enough.

1.6 Inclusion–Exclusion

Finite additivity directly handles disjoint events. If A and B overlap, adding $P(A)$ and $P(B)$ counts their intersection twice.

Proposition 2: Inclusion–exclusion for two events

For any events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof. Split $A \cup B$ into the disjoint events A and $B \cap A^c$:

$$P(A \cup B) = P(A) + P(B \cap A^c).$$

The event B is itself the disjoint union of $A \cap B$ and $A^c \cap B$. Thus $P(B \cap A^c) = P(B) - P(A \cap B)$, which gives the formula. $\square$

Example 6: Overlapping die events

Let $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$. Then $P(A) = P(B) = 1/2$ and $P(A \cap B) = 1/3$. Therefore

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = \frac{2}{3},$$

in agreement with $A \cup B = \{2, 4, 5, 6\}$.

Applying the two-event formula twice gives the three-event version:

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C). \end{aligned}$$

The final term restores outcomes in all three events: they were added three times and subtracted three times.

For n events, the same pattern continues:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{\emptyset \neq I \subseteq \{1, \dots, n\}} (-1)^{|I|+1} P\left(\bigcap_{i \in I} A_i\right).$$

To see why, consider an outcome contained in exactly k of the events. It is counted $\binom{k}{j}$ times among the intersections of size j , so its total coefficient on the right is $\sum_{j=1}^k (-1)^{j+1} \binom{k}{j} = 1$. An outcome in none of the events contributes zero to both sides.

1.7 Bounds for Unions and Intersections

Sometimes the probabilities of intersections are unknown. We can still bound the probability of a union.

Proposition 3: Union bound (Boole's inequality)

For events $A_1, \dots, A_n$,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

Proof. For two events, inclusion–exclusion gives $P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$ because $P(A_1 \cap A_2) \geq 0$. Apply the same two-event inequality to $(A_1 \cup \dots \cup A_n)$ and A_{n+1} . Starting with $P(A_1) \leq P(A_1)$, induction proves the claim. $\square$

For a countable sequence, the same inequality follows by applying the finite bound to the first n events and taking the limit.

Proposition 4: Bonferroni's inequality

For events $A_1, \dots, A_n$,

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n - 1).$$

Proof. The second n -event form of De Morgan's law turns the complement of an intersection into a union of complements. Then the union bound gives

$$\begin{aligned} P\left(\bigcap_{i=1}^n A_i\right) &= 1 - P\left(\bigcup_{i=1}^n A_i^c\right) \\ &\geq 1 - \sum_{i=1}^n P(A_i^c) \\ &= 1 - \sum_{i=1}^n (1 - P(A_i)) = \sum_{i=1}^n P(A_i) - (n - 1). \end{aligned}$$

$\square$

1.8 A Randomized Identity Test

Probability can quantify the error of a faster algorithm. Suppose F and G are polynomials of degree at most d , and we want to test whether $F(x) = G(x)$ for every x . Choose a value R uniformly from a finite set S , evaluate both polynomials at R , and report them different if the values disagree.

If $F = G$, the test is always correct. If $F \neq G$, the nonzero polynomial $H = F - G$ has at most d roots. A false report of equality occurs only when the chosen R is one of those roots. Consequently, when $F \neq G$, the probability over the choice of R satisfies

$$P(F(R) = G(R)) \leq \frac{d}{|S|}.$$

For $S = \{1, \dots, 100d\}$ with $d \geq 1$, this bound is $1/100$. The error is *one-sided*: finding a disagreement proves the polynomials are different, while a single agreement does not prove they are equal.

1.9 Odds

For an event A with $P(A) < 1$, its *odds in favor* are

$$O(A) = \frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}.$$

Conversely, $P(A) = O(A)/(1 + O(A))$. For a fair die, the odds in favor of rolling a six are $(1/6)/(5/6) = 1/5$, often stated as one to five. This is different from saying that the probability is $1/5$; the probability is $1/6$.